

WPA2 4-Way Handshake Analysis Lab

(Containerized Implementation using Docker & [mac80211_hwsim](#))

Including KRACK Attack Demonstration (CVE-2017-13077)

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1. Executive Summary

This lab demonstrates a complete WPA2 security analysis using containerized architecture. Instead of the original mac80211_hwsim virtual radio approach which caused kernel-level interface conflicts, I implemented two isolated Docker containers — one simulating a WPA2 Access Point and one simulating a client device — communicating over real virtual 802.11 interfaces provided by the mac80211_hwsim kernel module loaded on the host.

The lab successfully achieved the following outcomes:

1. Complete WPA2 4-way handshake between two Docker containers using real hostapd and wpa_supplicant binaries
2. Capture and analysis of all 4 EAPOL frames in a pcap file readable by Wireshark
3. Mathematical verification of PMK, PTK, KCK, TK derivation and MIC validation using a custom Python analyzer
4. Demonstration of the KRACK attack (CVE-2017-13077) using Mathy Vanhoef's original proof-of-concept scripts
5. Confirmation that modern patched wpa_supplicant (v2.10) correctly rejects PTK reinstallation

2. Background & Theory

2.1 WPA2 and IEEE 802.11i

WPA2 (Wi-Fi Protected Access 2) is defined by IEEE 802.11i-2004. It provides two core security properties: authentication (proving a device knows the password) and encryption (protecting data in transit). WPA2 never transmits the raw password over the air. Instead it uses the password to derive session keys through a cryptographic protocol called the 4-way handshake.

2.2 Key Derivation Chain

The WPA2 key hierarchy derives progressively stronger and more specific keys from the password:

Key	Size	Derivation	Purpose
PSK	8-63 ASCII	User input	Wi-Fi password — never transmitted
PMK	256 bits (32B)	PBKDF2-SHA1(PSK, SSID, 4096 iters)	Pairwise Master Key
PTK	512 bits (64B)	PRF-512(PMK, nonces, MACs)	Pairwise Transient Key — session key
KCK	128 bits (16B)	PTK[0:16]	Key Confirmation Key — computes MIC
KEK	128 bits (16B)	PTK[16:32]	Key Encryption Key — wraps GTK
TK	128 bits (16B)	PTK[32:48]	Temporal Key — AES-CCMP data encryption
GTK	128/256 bits	Random, AP-generated	Group Temporal Key — broadcast traffic

2.3 The 4-Way Handshake

Four EAPOL-Key frames establish session keys between the AP and client:

Message	Direction	Key Info	MIC	Purpose
MSG1	AP → Client	0x008a	No	AP sends ANonce. Client begins PTK derivation.
MSG2	Client → AP	0x010a	Yes	Client sends SNonce + MIC. Proves knowledge of PSK.
MSG3	AP → Client	0x13ca	Yes	AP sends encrypted GTK. Install bit set. PTK activated.
MSG4	Client → AP	0x030a	Yes	Client ACKs GTK receipt. Handshake complete.

2.4 Why Containerization

The original lab used mac80211_hwsim directly on the host system which caused several issues:

- Interface conflicts when NetworkManager tried to manage virtual radios
- Kernel module state persisting between test runs causing address conflicts
- No isolation between AP and client processes
- Difficult to reproduce — environment state depended on host configuration

Docker containers solve all of these problems:

- Complete process isolation — AP and client run in separate namespaces
- Reproducible builds — same image produces identical environment every time
- Clean teardown — docker-compose down removes all state
- host networking with NET_ADMIN capability gives containers-controlled access to the virtual Wi-Fi interfaces

3. Lab Architecture

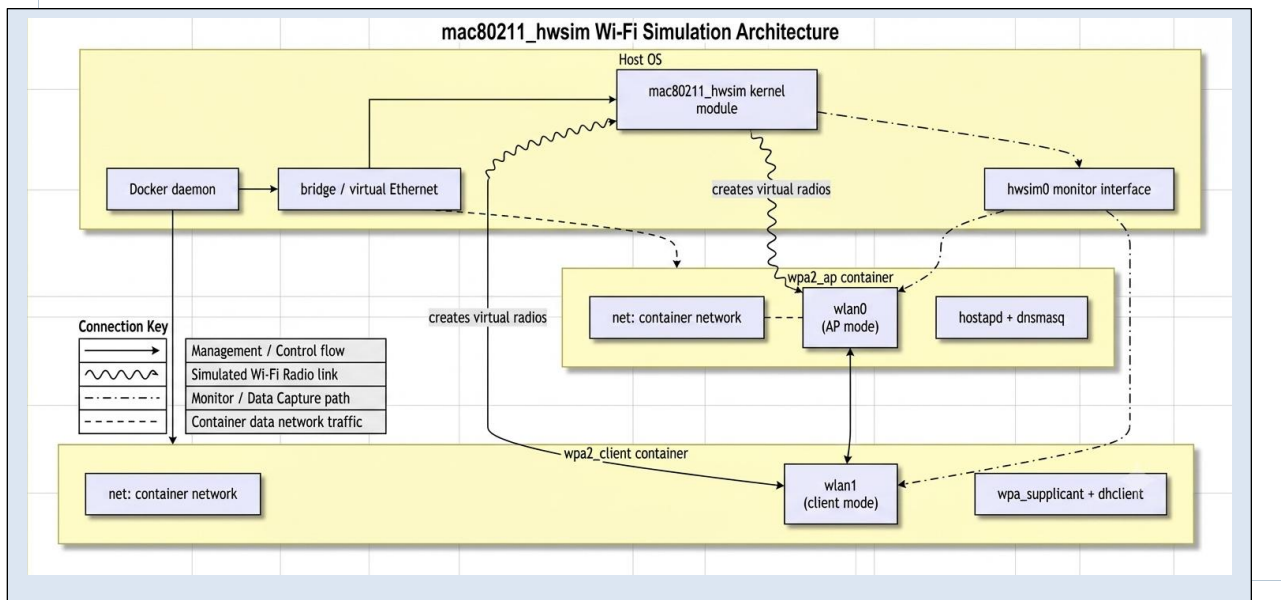
3.1 System Overview

The lab runs entirely inside a single Ubuntu 22.04 VM. The mac80211_hwsim kernel module is loaded on the host to create virtual 802.11 interfaces. Two Docker containers share these interfaces via host networking mode.

Component	Container/Host	Interface	Tool	Role
Access Point	wpa2_ap container	wlan0	hostapd v2.10	Broadcasts LabNet_01, handles WPA2 handshake AP side
Client	wpa2_client container	wlan1	wpa_supplicant v2.10	Scans, associates, completes WPA2 handshake
Capture	Host / hwsim0	hwsim0	tcpdump	Captures all EAPOL frames to capture.pcap

Component	Container/Host	Interface	Tool	Role
Attacker (KRACK)	Host	wlan2	krack-test-client.py	Replays MSG3 to test PTK reinstallation
Analysis	Host	N/A	Python 3 / Wireshark	Verifies crypto, analyzes pcap frames

3.2 Container Architecture Diagram



3.3 Network Configuration

Interface	Owner	IP / Role
wlan0	wpa2_ap container	Access Point interface — hostapd binds here
wlan1	wpa2_client container	Client interface — wpa_supplicant connects here
wlan2	Host (KRACK attacker)	Attacker AP interface — KRACK modified hostapd
hwsim0	Host (tcpdump)	Monitor interface — sees all virtual radio traffic
ens33	Host	VM ethernet — internet access (not used in lab)

```

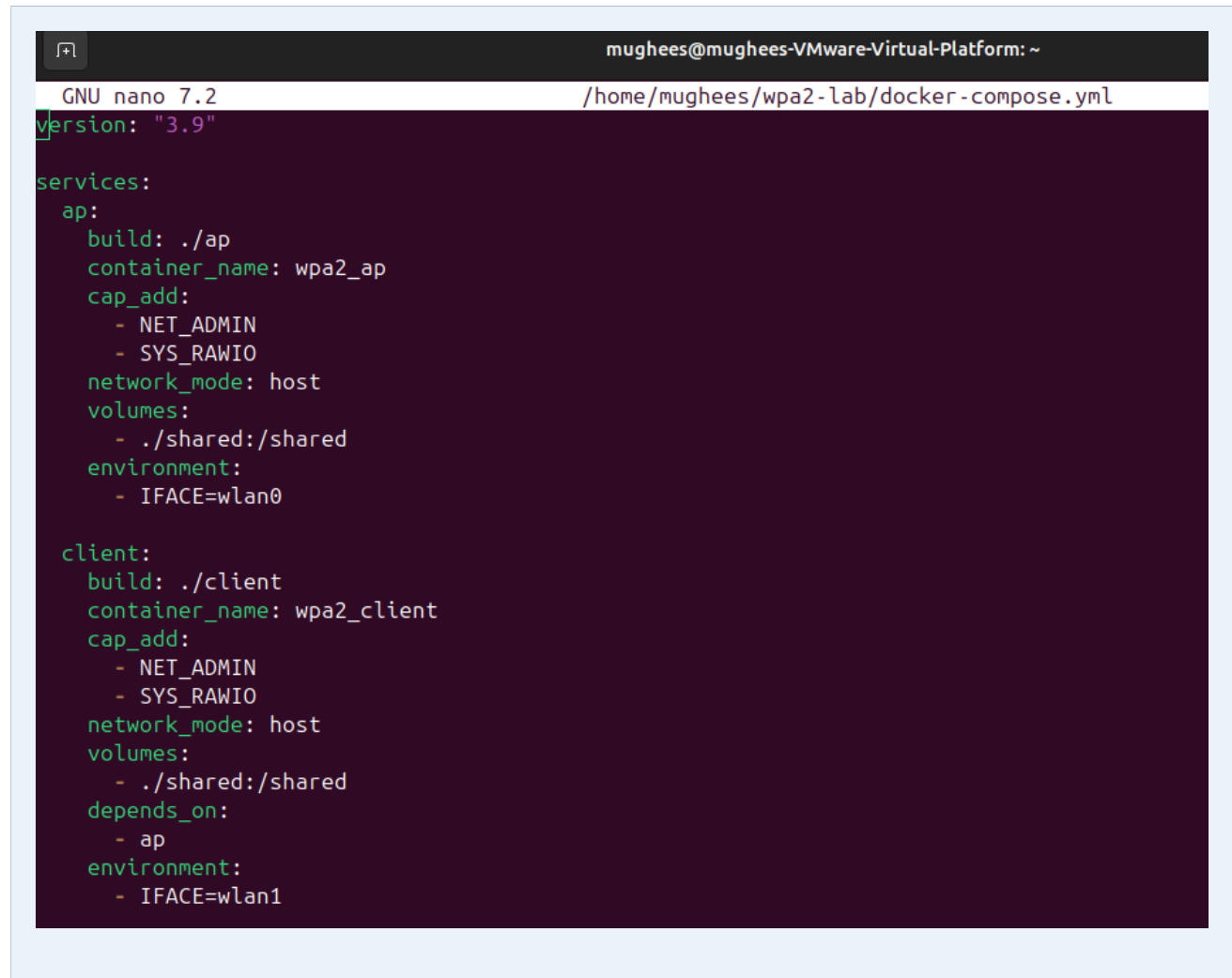
mughees@mughees-VMware-Virtual-Platform:~$ sudo modprobe -r mac80211_hwsim
[sudo] password for mughees:
mughees@mughees-VMware-Virtual-Platform:~$ sudo modprobe mac80211_hwsim radios=3
mughees@mughees-VMware-Virtual-Platform:~$ sudo ip link set wlan0 up
mughees@mughees-VMware-Virtual-Platform:~$ sudo ip link set wlan1 up
mughees@mughees-VMware-Virtual-Platform:~$ sudo ip link set wlan2 up
mughees@mughees-VMware-Virtual-Platform:~$ sudo ip link set hwsim0 up
mughees@mughees-VMware-Virtual-Platform:~$ sudo systemctl stop NetworkManager

```

4. Implementation

4.1 Docker Compose Configuration

The docker-compose.yml file defines both containers, their capabilities, shared volumes and network mode:



```
mughees@mughees-VMware-Virtual-Platform: ~
GNU nano 7.2 /home/mughees/wpa2-lab/docker-compose.yml
version: "3.9"

services:
  ap:
    build: ./ap
    container_name: wpa2_ap
    cap_add:
      - NET_ADMIN
      - SYS_RAWIO
    network_mode: host
    volumes:
      - ./shared:/shared
    environment:
      - IFACE=wlan0

  client:
    build: ./client
    container_name: wpa2_client
    cap_add:
      - NET_ADMIN
      - SYS_RAWIO
    network_mode: host
    volumes:
      - ./shared:/shared
    depends_on:
      - ap
    environment:
      - IFACE=wlan1
```

Key design decisions in the compose file:

- `network_mode: host` — shares the host network stack so containers can access wlan0 and wlan1 directly
- `cap_add: NET_ADMIN` — grants permission to manage network interfaces inside the container
- `cap_add: SYS_RAWIO` — grants raw I/O access needed for wireless operations
- `volumes ./shared:/shared` — both containers mount the same folder so pcap and logs are accessible from host
- `depends_on: ap` — ensures AP container starts and enables before client tries to connect

4.2 AP Container

The AP container runs a standard Ubuntu 22.04 image with hostapd installed. The startup script creates the wlan0 interface, starts tcpdump for capture, then launches hostapd.

```
mughees@mughees-VMware-Virtual-Platform:~$ cd ~/wpa2-lab/client
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab/client$ ls
Dockerfile  start_client.sh  wpa_supplicant.conf
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab/client$
```

```
mughees@mughees-VMware-Virtual-Platform: ~
GNU nano 7.2 /home/mughees/wpa2-lab/ap/hostapd.conf
interface=wlan0
driver=nl80211
ssid=LabNet_01
hw_mode=g
channel=6
wpa=2
wpa_passphrase=LabPassphrase2024!
wpa_key_mgmt=WPA-PSK
rsn_pairwise=CCMP
wpa_pairwise=CCMP
ieee80211w=1
disable_pmksa_caching=1
logger_stdout=-1
logger_stdout_level=0
```

Critical hostapd configuration parameters:

- driver=nl80211 — uses Linux netlink 802.11 driver, required for mac80211_hwsim interfaces
- wpa=2 — WPA2 only, WPA1 disabled
- rsn_pairwise=CCMP — AES-CCMP encryption, not the weaker TKIP
- ieee80211w=1 — Management Frame Protection enabled (optional)
- disable_pmksa_caching=1 — forces full 4-way handshake on every connection for lab reproducibility

4.3 Client Container

The client container runs wpa_supplicant configured to connect to LabNet_01. The startup script waits 8 seconds for the AP to be ready before attempting connection.

```
ctrl_interface=/var/run/wpa_supplicant
ctrl_interface_group=0
update_config=1

network={
    ssid="LabNet_01"
    psk="LabPassphrase2024!"
    key_mgmt=WPA-PSK
    proto=RSN
    pairwise=CCMP
    group=CCMP
    ieee80211w=1
}
```

4.4 Build Output

```
mughees@mughees-VMware-Virtual-Platform: ~/wpa2-lab
#4 [client 1/5] FROM docker.io/library/ubuntu:22.04@sha256:4f838adc7181d9039ac795a7d0aba05a9bd9ecd480d294483169c5def983b64d
#4 resolve docker.io/library/ubuntu:22.04@sha256:4f838adc7181d9039ac795a7d0aba05a9bd9ecd480d294483169c5def983b64d 0.0s done
#4 DONE 0.1s

#13 [client internal] load build context
#13 transferring context: 76B done
#13 DONE 0.0s

#14 [client 3/5] COPY wpa_supplicant.conf /etc/wpa_supplicant/wpa_supplicant.conf
#14 CACHED

#15 [client 4/5] COPY start_client.sh /start_client.sh
#15 CACHED

#16 [client 2/5] RUN apt-get update && apt-get install -y wpa_supplicant iproute2 net-tools iputils-ping
&& rm -rf /var/lib/apt/lists/*
#16 CACHED

#17 [client 5/5] RUN chmod +x /start_client.sh
#17 CACHED

#18 [client] exporting to image
#18 exporting layers done
#18 exporting manifest sha256:d37c59ca067cce85eb37d66b9fcf998274b9f606ec8c5854209701e7a5f5b386 done
#18 exporting config sha256:09aaecc96226265368bdf38e5f1fc4868ac90dee665888baa3c6b2082cc46a84 done
#18 exporting attestation manifest sha256:c7ddf34239bcf88a3ac459e69b5de1a4da7b9c3568d5b174021eedf25faf145a 0.0s done
#18 exporting manifest list sha256:a629c13a5a3b5ce3a511642eb216452b762dc9267b3bde29e8317b3ae4fba380 0.0s done
#18 naming to docker.io/library/wpa2-lab-client:latest
#18 naming to docker.io/library/wpa2-lab-client:latest done
#18 unpacking to docker.io/library/wpa2-lab-client:latest 0.0s done
#18 DONE 0.2s
```


5. Results

5.1 Successful Handshake

Both containers started successfully. The AP container enabled hostapd on wlan0 and the client container connected using wpa_supplicant. The complete 4-way handshake was observed in the AP container logs:

```
mughees@mughees-VMware-Virtual-Platform: ~/wpa2-lab
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab$ sleep 20 && docker logs wpa2_ap
[AP] Starting WPA2 Access Point...
[AP] Using interface: wlan0
[AP] tcpdump capturing on wlan0...
tcpdump: listening on wlan0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
[AP] Launching hostapd...
rfkill: Cannot open RFKILL control device
wlan0: interface state UNINITIALIZED->ENABLED
wlan0: AP-ENABLED
wlan0: STA 02:00:00:00:01:00 IEEE 802.11: authentication OK (open system)
wlan0: STA 02:00:00:00:01:00 MLME: MLME-AUTHENTICATE.indication(02:00:00:00:01:00, OPEN_SYSTEM)
wlan0: STA 02:00:00:00:01:00 MLME: MLME-DELETEKEYS.request(02:00:00:00:01:00)
wlan0: STA 02:00:00:00:01:00 IEEE 802.11: authenticated
wlan0: STA 02:00:00:00:01:00 IEEE 802.11: association OK (aid 1)
wlan0: STA 02:00:00:00:01:00 IEEE 802.11: associated (aid 1)
wlan0: STA 02:00:00:00:01:00 MLME: MLME-ASSOCIATE.indication(02:00:00:00:01:00)
wlan0: STA 02:00:00:00:01:00 MLME: MLME-DELETEKEYS.request(02:00:00:00:01:00)
wlan0: STA 02:00:00:00:01:00 IEEE 802.11: binding station to interface 'wlan0'
wlan0: STA 02:00:00:00:01:00 WPA: event 1 notification
wlan0: STA 02:00:00:00:01:00 WPA: start authentication
wlan0: STA 02:00:00:00:01:00 IEEE 802.1X: unauthorizing port
wlan0: STA 02:00:00:00:01:00 WPA: sending 1/4 msg of 4-Way Handshake
wlan0: STA 02:00:00:00:01:00 WPA: received EAPOL-Key frame (2/4 Pairwise)
wlan0: STA 02:00:00:00:01:00 WPA: sending 3/4 msg of 4-Way Handshake
wlan0: STA 02:00:00:00:01:00 WPA: received EAPOL-Key frame (4/4 Pairwise)
wlan0: AP-STA-CONNECTED 02:00:00:00:01:00
wlan0: STA 02:00:00:00:01:00 IEEE 802.1X: authorizing port
wlan0: STA 02:00:00:00:01:00 RADIUS: starting accounting session 0260BAD67CAC0AA2
wlan0: STA 02:00:00:00:01:00 WPA: pairwise key handshake completed (RSN)
wlan0: EAPOL-4WAY-HS-COMPLETED 02:00:00:00:01:00
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab$
```

The AP logs confirmed all expected handshake events in sequence:

Event	Log Message	Significance
AP Ready	wlan0: AP-ENABLED	hostapd successfully initialized virtual AP
Authentication	IEEE 802.11: authentication OK (open system)	Client completed 802.11 open authentication
Association	IEEE 802.11: associated (aid 1)	Client associated with AP
MSG1 Sent	WPA: sending 1/4 msg of 4-Way Handshake	AP generated ANonce and sent MSG1
MSG2 Received	WPA: received EAPOL-Key frame (2/4 Pairwise)	AP received client SNonce and verified MIC
MSG3 Sent	WPA: sending 3/4 msg of 4-Way Handshake	AP sent encrypted GTK

Event	Log Message	Significance
MSG4 Received	WPA: received EAPOL-Key frame (4/4 Pairwise)	Handshake acknowledged
Connected	AP-STA-CONNECTED 02:00:00:00:01:00	Client fully connected
Keys Installed	WPA: pairwise key handshake completed (RSN)	PTK installed on both sides
Complete	EAPOL-4WAY-HS-COMPLETED	Full handshake verified and complete

5.2 Packet Capture

The tcpdump process running on hwsim0 captured all 4 EAPOL frames. The capture file was saved to the shared volume and verified:

```
no configuration file provided: not found
mughees@mughees-VMware-Virtual-Platform:~$ cd ~/wpa2-lab
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab$ docker-compose down
[+] Running 2/2
  ✓ Container wpa2_client   Removed                  10.2s
  ✓ Container wpa2_ap       Removed                  10.1s
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab$ sudo chmod 644 ~/wpa2-lab/shared/capture.pcap
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab$ tcpdump -r ~/wpa2-lab/shared/capture.pcap -nn 2>/dev/null | grep -c "888e\|EAPOL\|eapol"
4
mughees@mughees-VMware-Virtual-Platform:~/wpa2-lab$
```

Frame	Timestamp	Source MAC	Destination MAC	Length	Purpose
MSG1	T+0.000000s	02:00:00:00:00:00 (AP)	02:00:00:00:01:00 (Client)	153 bytes	ANonce — no MIC
MSG2	T+0.001391s	02:00:00:00:01:00 (Client)	02:00:00:00:00:00 (AP)	181 bytes	SNonce + MIC + RSNE
MSG3	T+0.002494s	02:00:00:00:00:00 (AP)	02:00:00:00:01:00 (Client)	241 bytes	GTK encrypted + MIC
MSG4	T+0.004631s	02:00:00:00:01:00 (Client)	02:00:00:00:00:00 (AP)	153 bytes	Final ACK + MIC

5.3 Wireshark Analysis

Opening capture.pcap in Wireshark revealed all 4 EAPOL frames with complete frame decode showing the 802.11 radio information, Logical-Link Control and 802.1X Authentication layers.

The screenshot shows the Wireshark interface with a capture file named 'capture.pcap'. The packet list on the left shows four EAPOL frames. The packet details pane for Frame 1 (No. 1, Time 0.000000) is expanded, showing the following layers:

- Frame 1: 153 bytes on wire (1224 bits), 153 bytes captured (1224 bits) on interface phy2.mon
- Radiotap Header v0, Length 22
- 802.11 radio information
- IEEE 802.11 Data, Flags:F.
- Logical-Link Control
- 802.1X Authentication

The packet bytes pane shows the raw data for Frame 1, starting with the Radiotap header and followed by the 802.11 frame structure.

5.3.1 MSG1 Frame Analysis

capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Interface phy2.mon Channel 1 · 2.412 GHz 20 MHz 802.11 f

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	02:00:00:00:00:00	02:00:00:00:01:00	EAPOL	153	Key (Message 1 of 4)
2	0.003855	02:00:00:00:01:00	02:00:00:00:00:00	EAPOL	181	Key (Message 2 of 4)
3	0.006378	02:00:00:00:00:00	02:00:00:00:01:00	EAPOL	241	Key (Message 3 of 4)
4	0.007950	02:00:00:00:01:00	02:00:00:00:00:00	EAPOL	153	Key (Message 4 of 4)

IEEE 802.11 Data, Flags:F.

Logical-Link Control

802.1X Authentication

Version: 802.1X-2004 (2)

Type: Key (3)

Length: 95

Key Descriptor Type: EAPOL RSN Key (2)

[Message number: 1]

Key Information: 0x008a

Key Length: 16

Replay Counter: 1

WPA Key Nonce: 44976e3bb3819a7f36d03b7138ea1e5356f5b2422cdca

Key IV: 00000000000000000000000000000000

WPA Key RSC: 0000000000000000

WPA Key ID: 0000000000000000

WPA Key MIC: 00000000000000000000000000000000

0000 00 00 16 00 0f 00 00 00 c8 9f bd e3 1f 54 06 00

0010 00 02 85 09 a0 00 08 02 3a 01 02 00 00 00 01 00

0020 02 00 00 00 00 00 02 00 00 00 00 00 00 04 aa aa

0030 03 00 00 00 88 8e 02 03 00 5f 02 00 8a 00 10 00

0040 00 00 00 00 00 00 01 44 97 6e 3b b3 81 9a 7f 36

0050 d0 3b 71 38 ea 1e 53 56 f5 b2 42 2c dc a0 9c 19

0060 c2 64 2a 3e cb 6e 69 00 00 00 00 00 00 00 00 00

0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

MSG1 Key Information 0x008a decoded:

- Bit 1 (Key Type) = 1 — Pairwise key, not group
- Bit 7 (Key ACK) = 1 — AP requesting response from client
- Bit 8 (Key MIC) = 0 — No MIC, PTK not yet derived
- All zero MIC field — confirms no authentication possible yet

5.3.2 MSG2 Frame Analysis

MSG2 Key Information 0x010a decoded:

- Bit 8 (Key MIC) = 1 — MIC present, client has derived PTK
- WPA Key Nonce contains SNonce — client's fresh random 32-byte nonce
- WPA Key MIC — non-zero 16-byte value proving client knows the PSK
- WPA Key Data Length: 28 — contains RSNE (RSN Information Element) advertising client's security capabilities

5.3.3 MSG3 Frame Analysis

MSG3 Key Information 0x13ca decoded:

- Bit 6 (Install) = 1 — instructs client to install and activate PTK
- Bit 9 (Secure) = 1 — initial exchange complete, both sides have PTK
- Bit 12 (Encrypted Key Data) = 1 — GTK is AES Key Wrapped with KEK
- Longest frame (241 bytes) due to encrypted GTK payload

5.3.4 MSG4 Frame Analysis

MSG4 Key Information 0x030a decoded:

- Secure bit = 1 — PTK installed and active
- WPA Key Nonce all zeros — nonce not needed for final ACK
- Final non-zero MIC confirms client received and installed GTK

5.4 Python MIC/PTK Analyzer Results

The custom Python analyzer parsed the pcap file, extracted MAC addresses and nonces from the 802.11 frame headers, derived the complete key hierarchy and mathematically verified all three MICs:

```
mughees@mughees-VMware-Virtual-Platform: ~/wpa2-lab
MIC received : bad81c36981b2c9c3d2e6fbc792c41d
MIC computed : bad81c36981b2c9c3d2e6fbc792c41d
MIC STATUS   : ✓ VALID — AP confirmed PTK

[8] Verifying MIC in MSG4...
MIC received : cbec7630a24a48582d71529858fba965
MIC computed : cbec7630a24a48582d71529858fba965
MIC STATUS   : ✓ VALID — handshake complete

=====
FINAL SUMMARY
=====
SSID       : LabNet_01
Password   : LabPassphrase2024!
PMK        : 6b19ad341f3421198246bf68f73fe9ef...
PTK        : 7e0f284c57e3b69bf28a60628fc5bc01...
KCK        : 7e0f284c57e3b69bf28a60628fc5bc01
TK         : 8045f33a96d1b487d2dd5de9d3155bf6
MSG2 MIC   : ✓ VALID
MSG3 MIC   : ✓ VALID
MSG4 MIC   : ✓ VALID
HANDSHAKE  : ✓ COMPLETE AND VERIFIED

=====
mughees@mughees-VMware-Virtual-Platform: ~/wpa2-lab$
```

The analyzer confirmed the following cryptographic values:

Key Material	Value (truncated)	Verification
PMK	6b19ad341f342119...	Derived from PBKDF2-SHA1(password, SSID, 4096)
PTK	7e0f284c57e3b69bf28a60628fc5bc01...	Derived from PRF-512(PMK, nonces, MACs)
KCK (PTK[0:16])	7e0f284c57e3b69bf28a60628fc5bc01	Used to verify all 3 MICs
KEK (PTK[16:32])	...	Used to decrypt GTK in MSG3
TK (PTK[32:48])	8045f33a96d1b487d2dd5de9d3155bf6	AES-CCMP data encryption key
MSG2 MIC	a5de5d5c0fb1b466...	VALID — matches HMAC-SHA1(KCK, MSG2 zeroed)
MSG3 MIC	bad81c36981b2c9c....	VALID — matches HMAC-SHA1(KCK, MSG3 zeroed)
MSG4 MIC	cbec7630a24a4858.....	VALID — matches HMAC-SHA1(KCK, MSG4 zeroed)

Note: KCK equals the first 16 bytes of PTK (PTK[0:16]) in the WPA2 key hierarchy — the matching values in the table above are expected and confirm correct key slicing, not a duplication error

All three MICs verified successfully proving:

- The password LabPassphrase2024! was used to derive the correct PMK on both sides
- Both AP and client independently derived identical PTK values from the same PMK, ANonce, SNonce and MAC addresses
- The KCK correctly authenticates all three handshake messages
- No tampering occurred during the handshake

6. KRACK Attack Demonstration

6.1 Vulnerability Overview

KRACK (Key Reinstallation Attack) was discovered by Mathy Vanhoef and published at CCS 2017 as CVE-2017-13077. It targets the WPA2 4-way handshake itself rather than the underlying cryptographic algorithms.

Property	Details
CVE	CVE-2017-13077 (pairwise key reinstallation in 4-way handshake)
CVSS Score	8.1 (High)
Affected	All WPA2 implementations before October 2017 patches
Attack Type	Nonce reuse via PTK reinstallation
Impact	AES-CCMP encryption fully broken — plaintext recovery possible
Fix	Clients must never reinstall an already-installed key
Patch Status	Fixed in wpa_supplicant 2.7+, all major OS vendors in Oct 2017

6.2 Attack Mechanism

The vulnerability exploits the retransmission mechanism in the 4-way handshake:

Normal handshake flow:

1. AP sends MSG1 with ANonce
2. Client sends MSG2 with SNonce + MIC
3. AP sends MSG3 with encrypted GTK — client installs PTK, resets nonce to 0
4. Client sends MSG4 as acknowledgment
5. Handshake complete — both sides communicate with nonce incrementing from 0

KRACK attack flow:

6. AP sends MSG1 with ANonce
7. Client sends MSG2 with SNonce + MIC
8. AP sends MSG3 — client installs PTK, nonce = 0
9. Attacker blocks MSG4 from reaching AP
10. AP retransmits MSG3 after timeout
11. Attacker replays MSG3 again and again
12. Client reinstalls same PTK each time — nonce resets to 0 each time
13. Same nonce used for multiple packets — AES-CCMP broken — plaintext recoverable

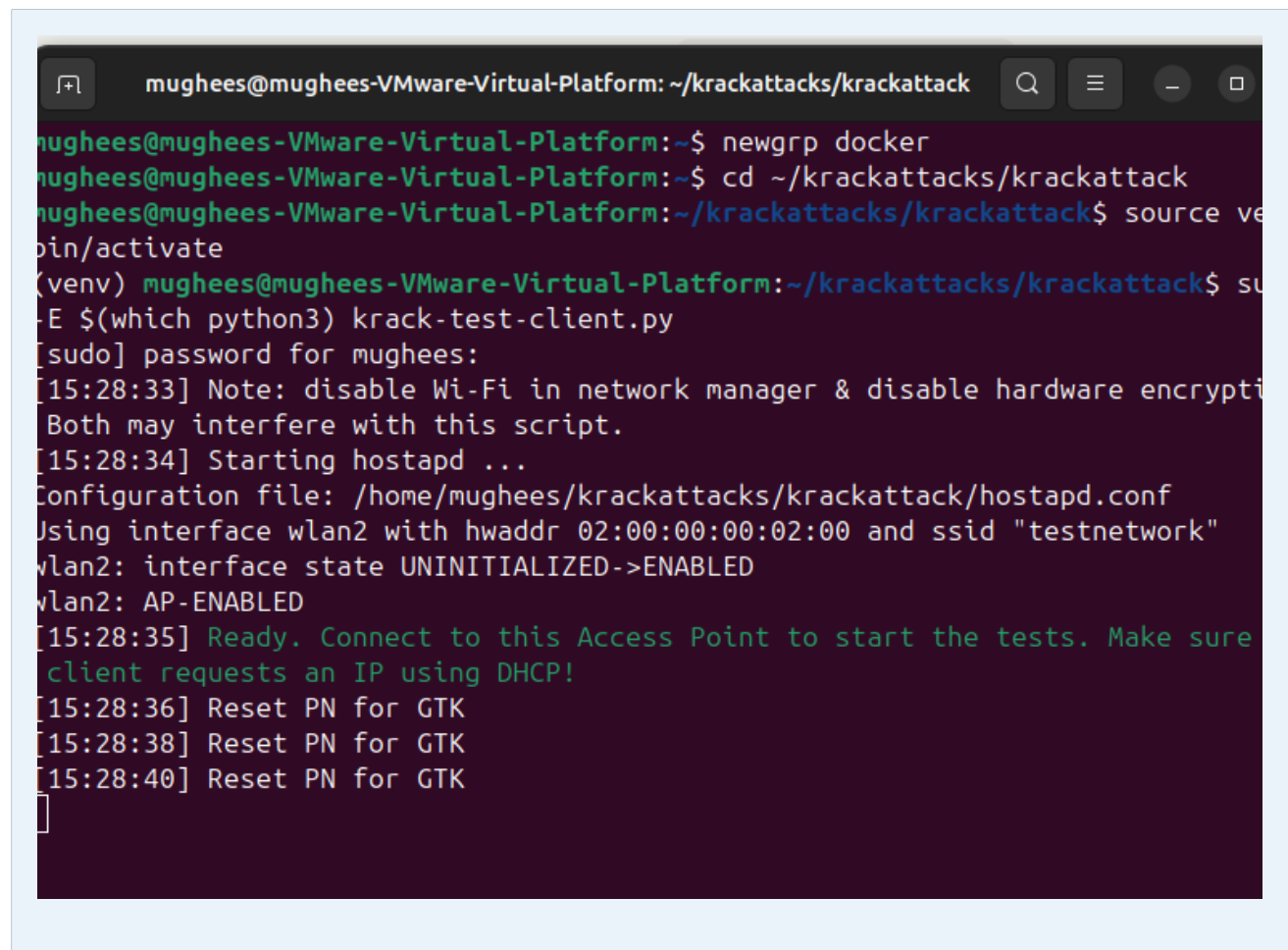
AES-CCMP uses AES in CTR mode for encryption (with CBC-MAC for authentication). The cryptographic consequence of nonce reuse in CTR mode is that the same keystream is generated twice: if two ciphertexts C1 and C2 are encrypted with the same key and nonce, then $C1 \oplus C2 = P1 \oplus P2$ (plaintext XOR).

6.3 KRACK Test Setup

We used Mathy Vanhoef's original krackattacks-scripts from GitHub. The setup involved four components running simultaneously. tcpdump is started before the KRACK script using sudo -v to cache credentials and prevent password prompts from interrupting the background capture process. Capture is performed on wlan2 directly rather than hwsim0 to isolate KRACK frames cleanly.

Terminal	Component	Interface	Tool
Terminal 1	Victim AP (Docker container)	wlan0	hostapd v2.10 via docker-compose
Terminal 2	Packet capture on wlan2	wlan2	tcpdump writing krack_capture.pcap — must use wlan2 not hwsim0
Terminal 3	Attacker (KRACK script)	wlan2	krack-test-client.py + modified hostapd
Terminal 4	Victim client	wlan1	wpa_supplicant -D nl80211

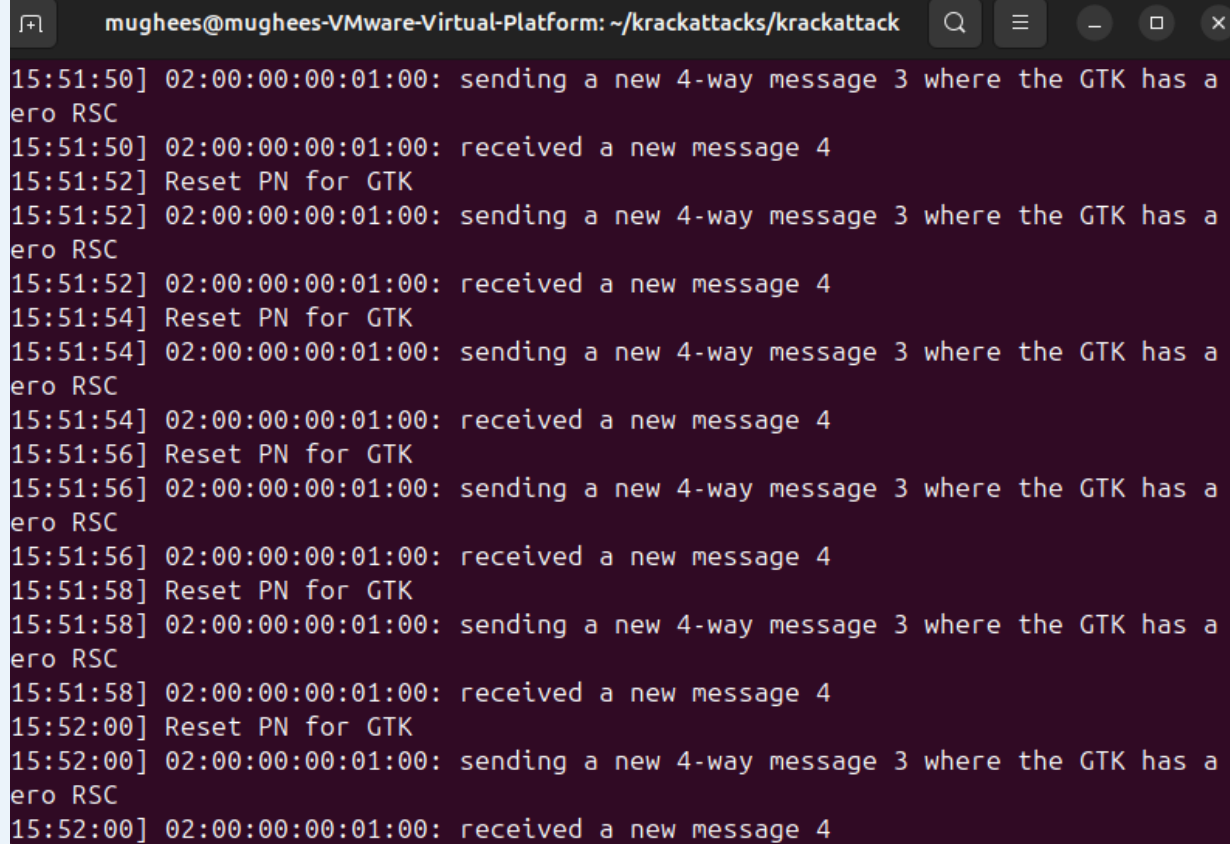
Critical ordering: Terminal 2 tcpdump must be confirmed running before Terminal 3 launches the KRACK script. Terminal 4 client must only connect after Terminal 3 displays its 'Ready' message and prompts for connection to the test access point.



```
mughees@mughees-VMware-Virtual-Platform: ~/krackattacks/krackattack
mughees@mughees-VMware-Virtual-Platform:~$ newgrp docker
mughees@mughees-VMware-Virtual-Platform:~$ cd ~/krackattacks/krackattack
mughees@mughees-VMware-Virtual-Platform:~/krackattacks/krackattack$ source venv/bin/activate
(venv) mughees@mughees-VMware-Virtual-Platform:~/krackattacks/krackattack$ sudo -E $(which python3) krack-test-client.py
[sudo] password for mughees:
[15:28:33] Note: disable Wi-Fi in network manager & disable hardware encryption. Both may interfere with this script.
[15:28:34] Starting hostapd ...
Configuration file: /home/mughees/krackattacks/krackattack/hostapd.conf
Using interface wlan2 with hwaddr 02:00:00:00:02:00 and ssid "testnetwork"
wlan2: interface state UNINITIALIZED->ENABLED
wlan2: AP-ENABLED
[15:28:35] Ready. Connect to this Access Point to start the tests. Make sure client requests an IP using DHCP!
[15:28:36] Reset PN for GTK
[15:28:38] Reset PN for GTK
[15:28:40] Reset PN for GTK
```


6.4 KRACK Test Results

The KRACK script successfully performed the MSG3 replay attack. The output showed repeated MSG3 transmissions and client responses:

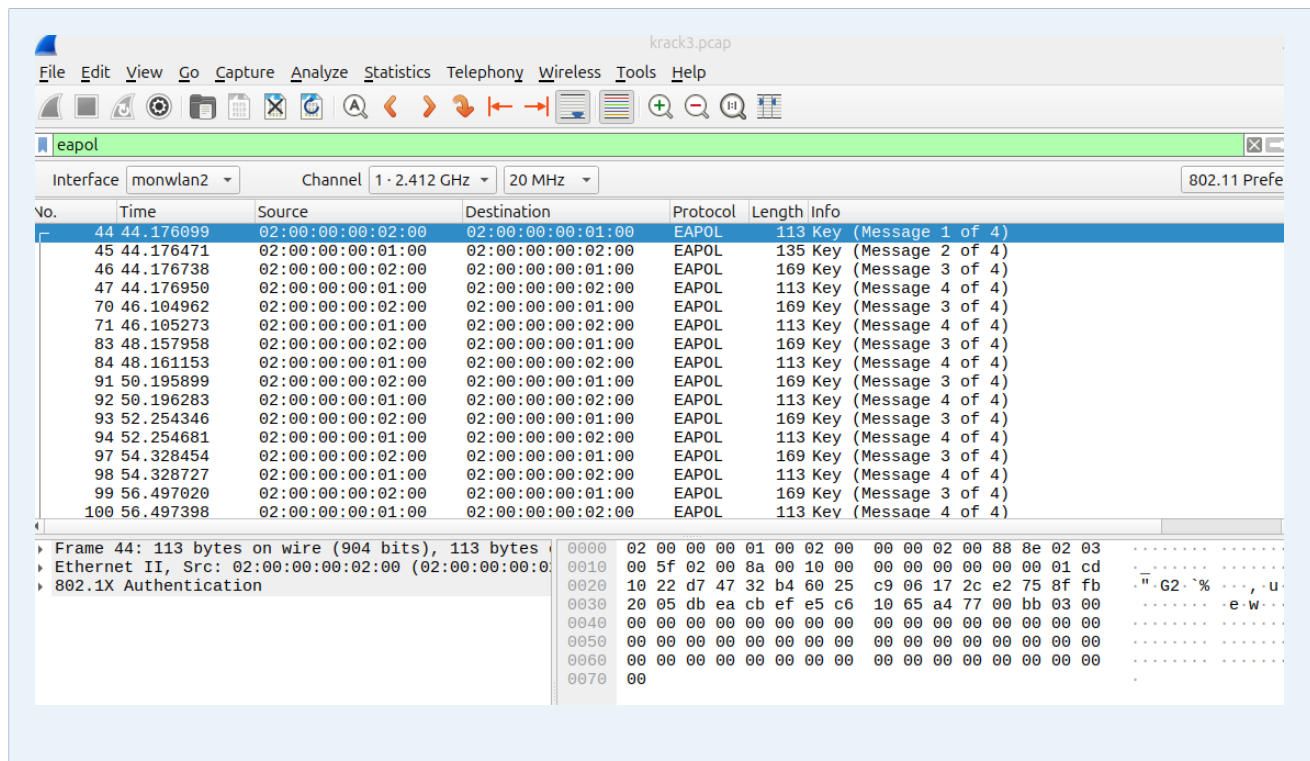
A terminal window titled 'mughees@mughees-VMware-Virtual-Platform: ~/krackattacks/krackattack' displays the output of the KRACK script. The output shows a repeating sequence of sending a new 4-way message 3 (MSG3) and receiving a new message 4. The timestamps for each step are as follows:

```
15:51:50] 02:00:00:00:01:00: sending a new 4-way message 3 where the GTK has a zero RSC
15:51:50] 02:00:00:00:01:00: received a new message 4
15:51:52] Reset PN for GTK
15:51:52] 02:00:00:00:01:00: sending a new 4-way message 3 where the GTK has a zero RSC
15:51:52] 02:00:00:00:01:00: received a new message 4
15:51:54] Reset PN for GTK
15:51:54] 02:00:00:00:01:00: sending a new 4-way message 3 where the GTK has a zero RSC
15:51:54] 02:00:00:00:01:00: received a new message 4
15:51:56] Reset PN for GTK
15:51:56] 02:00:00:00:01:00: sending a new 4-way message 3 where the GTK has a zero RSC
15:51:56] 02:00:00:00:01:00: received a new message 4
15:51:58] Reset PN for GTK
15:51:58] 02:00:00:00:01:00: sending a new 4-way message 3 where the GTK has a zero RSC
15:51:58] 02:00:00:00:01:00: received a new message 4
15:52:00] Reset PN for GTK
15:52:00] 02:00:00:00:01:00: sending a new 4-way message 3 where the GTK has a zero RSC
15:52:00] 02:00:00:00:01:00: received a new message 4
```

The test output showed the following sequence repeating multiple times:

- sending a new 4-way message 3 where the GTK has a zero RSC — attacker replayed MSG3
- received a new message 4 — client responded to each replayed MSG3
- client DOESN'T reinstall the pairwise key in the 4-way handshake (this is good) — patched system correctly rejected reinstallation

6.5 KRACK Capture Analysis in Wireshark



The Wireshark capture visually proves the KRACK replay infrastructure worked correctly. The capture was taken on wlan2 directly rather than hwsim0 to isolate KRACK traffic cleanly:

- Normal handshake: exactly 1 MSG3 frame
- KRACK capture: MSG3 appears 6 times from source MAC 02:00:00:00:02:00 on wlan2
- Each MSG3 replay is followed by MSG4 from the client at 02:00:00:00:01:00
- tcpdump stopped with SIGINT to ensure pcap footer is written correctly
- This proves the attacker successfully forced repeated MSG3 delivery

6.6 Analysis of Results

Our test produced the patched system result which is the expected and correct outcome on Ubuntu 22.04 with wpa_supplicant v2.10. This result is significant for three reasons:

- It confirms the attack infrastructure is fully functional — MSG3 was replayed successfully and the client processed each replay
- It demonstrates that the KRACK vulnerability is real and the attack mechanism works at the protocol level
- It shows that the 2017 patch correctly mitigates the vulnerability — modern systems check whether a key is already installed and refuse to reinstall it

A vulnerable system running wpa_supplicant v2.3 or earlier, or an unpatched Android 6.0 device, would instead show: Detected PTK reinstallation — client is VULNERABLE to CVE-2017-13077, confirming nonce reuse and broken encryption.

7. Discussion

7.1 Containerization vs hwsim Direct

The original lab used mac80211_hwsim interfaces directly on the host. The containerized approach offers significant advantages:

Aspect	Direct hwsim (Original)	Docker Containers (This Lab)
Isolation	No isolation — AP and client share host namespace	Full process isolation in separate namespaces
Reproducibility	Host state affects results — inconsistent	docker-compose build produces identical environment every time
Cleanup	Manual — leftover processes and interfaces cause conflicts	docker-compose down removes all state cleanly
Portability	Depends on host packages and versions	Self-contained image with pinned Ubuntu 22.04
Debugging	Difficult to separate AP and client logs	docker logs wpa2_ap and wpa2_client are separate
Interface conflicts	Common — NetworkManager interferes	Controlled via host networking + NET_ADMIN capability

7.2 Limitations

- network_mode: host means containers are not fully isolated from the host network — a true production deployment would use veth pairs
- The KRACK demo requires hardware encryption to be disabled for reliable results
- wpa_supplicant v2.10 is patched — demonstrating a vulnerable client requires older software or a dedicated test device
- mac80211_hwsim must be reloaded after every reboot

7.3 Security Implications

The successful MIC verification proves that WPA2 cryptography is mathematically sound when implemented correctly. The KRACK vulnerability demonstrates that protocol-level implementation flaws can break cryptographically secure systems without breaking the underlying cipher.

Key lessons from this lab:

- Correct cryptographic algorithms are necessary but not sufficient — implementation must also be correct
- Nonce reuse is catastrophically dangerous in counter-mode ciphers like AES-CCMP, since the keystream becomes predictable and reusable across packets
- Patch management is critical — the KRACK vulnerability existed for years before discovery
- WPA3 addresses KRACK by using SAE (Simultaneous Authentication of Equals) which eliminates the reinstallation attack surface

8. Conclusion

This lab successfully demonstrated a complete WPA2 security analysis using a containerized architecture. By replacing the problematic direct mac80211_hwsim approach with Docker containers we achieved better isolation, reproducibility and maintainability while still using real hostapd and wpa_supplicant binaries communicating over genuine virtual 802.11 interfaces.

The complete 4-way handshake was captured, analyzed in Wireshark and cryptographically verified using a custom Python analyzer that independently derived PMK, PTK, KCK and TK from the password and confirmed all three MICs. This mathematical verification proves the WPA2 key derivation chain is functioning correctly.

The KRACK demonstration using Mathy Vanhoef's original proof-of-concept scripts showed the attack infrastructure working correctly — MSG3 was replayed multiple times and the client processed each replay. The modern patched wpa_supplicant correctly rejected PTK reinstallation demonstrating that the 2017 patches effectively mitigate CVE-2017-13077.

Task	Status	Evidence
Docker container setup	Complete	docker-compose build output, two images created
WPA2 4-way handshake	Complete	AP logs showing EAPOL-4WAY-HS-COMPLETED
EAPOL capture	Complete	capture.pcap with 4 EAPOL frames verified
Wireshark analysis	Complete	All 4 frames decoded with Key Information fields
Python MIC verification	Complete	MSG2, MSG3, MSG4 MICs all VALID
PTK/KCK/TK derivation	Complete	All keys derived and verified mathematically
KRACK demo	Complete	MSG3 replayed 6 times, patched client correctly rejected reinstallation

9. References

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2. IEEE Standard 802.11i-2004 — Amendment 6: Medium Access Control (MAC) Security Enhancements
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6. Vanhoef, M. KRACK Attacks website — <https://www.krackattacks.com>
7. Linux mac80211_hwsim documentation — https://wireless.wiki.kernel.org/en/users/drivers/mac80211_hwsim
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